

Credit Risk Portfolio Review — Part 1

10,000 funded SMB loans | 11 April 2018 to 18 December 2025

Interactive dashboard: <https://datastudio.google.com/s/taLG52JGGKY>

Executive Summary

Portfolio performance has become more costly even though loss frequency is not at its historical peak: the 2025 vintage recorded the highest average net loss per funded loan at \$8,987 and the highest average loss severity at \$38,250, while its 23.76% loss rate remained below the 26.71% peak in 2023. Risk is also concentrated in specific segments and in the loss tail: IND_019 and IND_043 combine elevated loss rates with severity near \$52,000, and the largest 10% of positive-loss cases account for 36.69% of positive losses; these results should be read alongside material duplicate structure, outcome conflicts and sentinel-coded missingness.

Methodology

Sample. 10,000 funded SMB loans originated from 11 April 2018 to 18 December 2025.

Data source. YHP_credit_assessment_DS.sqlite, comprising a loans table with loan_id, received_date, is_loss and loss_amount, and a loan_features table with 99 anonymised predictors.

Tools and workflow. Repeatable transformations were built in dbt on DuckDB using staging, intermediate and mart layers; the full build runs 124 tests, all passing. Looker Studio is used for the interactive visualisation layer.

Measures. The supplied brief defines loss_amount as realised loss in \$USD. Loss rate is based on is_loss; average loss severity in the vintage mart is the mean loss_amount among flagged loss cases; average net loss per funded loan is total recorded loss, including negative recoveries, divided by funded loans.

Portfolio baseline. Across the full sample, 2,119 loans are flagged as losses (21.19%). Total net recorded loss is \$ 68.53 million, equivalent to \$6,853 per funded loan; 2,232 loans have a positive loss amount, with average positive-loss severity of \$30,872.

Key Findings

Finding 1 — Economic loss per loan increased as severity rose

Loss frequency and loss size are not correlated across vintages. The observed loss rate peaked at 26.71% in 2023, whereas 2025 recorded a lower rate of 23.76%. However, 2025 had the highest average loss severity at \$38,250 and the highest average net loss per funded loan at \$8,987, 59% above the 2018 level of \$5,661. Over the latest 12-month window used in the mart, the loss rate was 23.46% across 1,573 loans compared with 21.91% across 1,698 loans in the preceding window, a rise of 1.55 percentage points(pp).

Loss Ratve v Loss Severity v Loss per Funded Loan

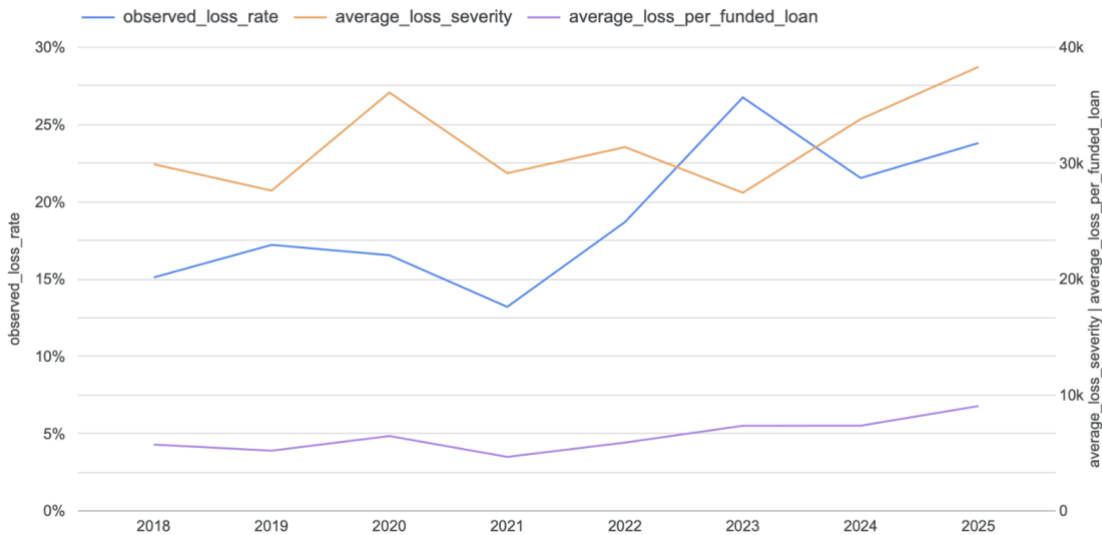


Figure 1. Annual loss rate and average net loss per funded loan

Business interpretation

Loss rate captures how often losses occur, while net loss per funded loan captures the economic effect of both frequency and realised loss amounts. A view based only on the decline from the 2023 loss-rate peak would therefore understate the worsening cost profile in 2025. Fourteen of the 23 largest recorded losses come from 2024 and 2025, although those vintages represent 32% of funded loans, providing additional evidence that recent deterioration is concentrated in severity as well as frequency.

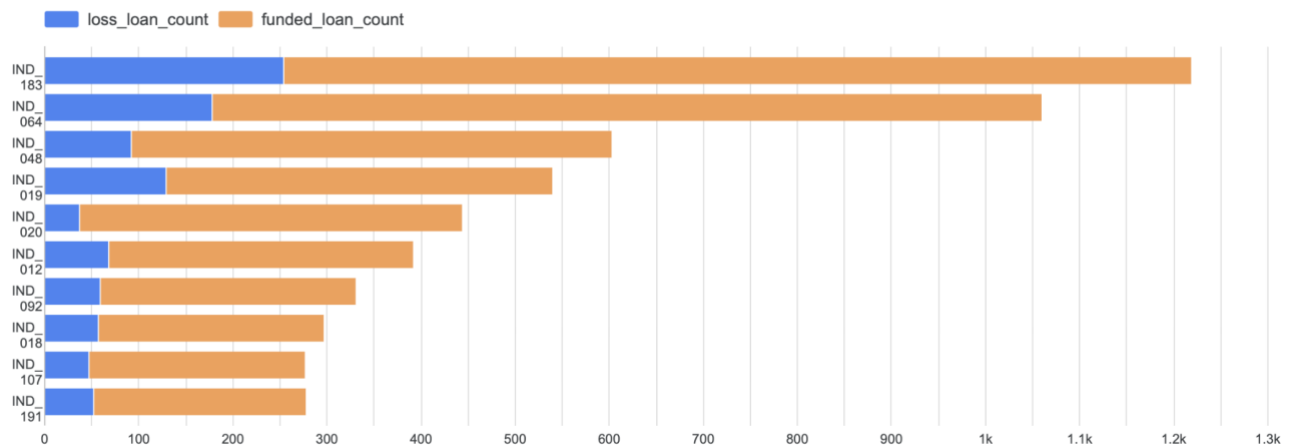
Limitations and practical context

2025 is the least seasoned vintage and has had less time to reach its eventual loss rate, so direct comparisons with older cohorts are incomplete. The pattern is not explained by seasoning alone because recent loss rates are non-monotonic (26.71% in 2023, 21.50% in 2024 and 23.76% in 2025). The data also contain no original balance, exposure-at-default or recovery fields, so the increase in severity cannot be attributed to larger facilities, weaker recoveries or another exposure effect. In practice, frequency controls such as underwriting thresholds should be assessed separately from severity controls such as exposure limits and collateral requirements.

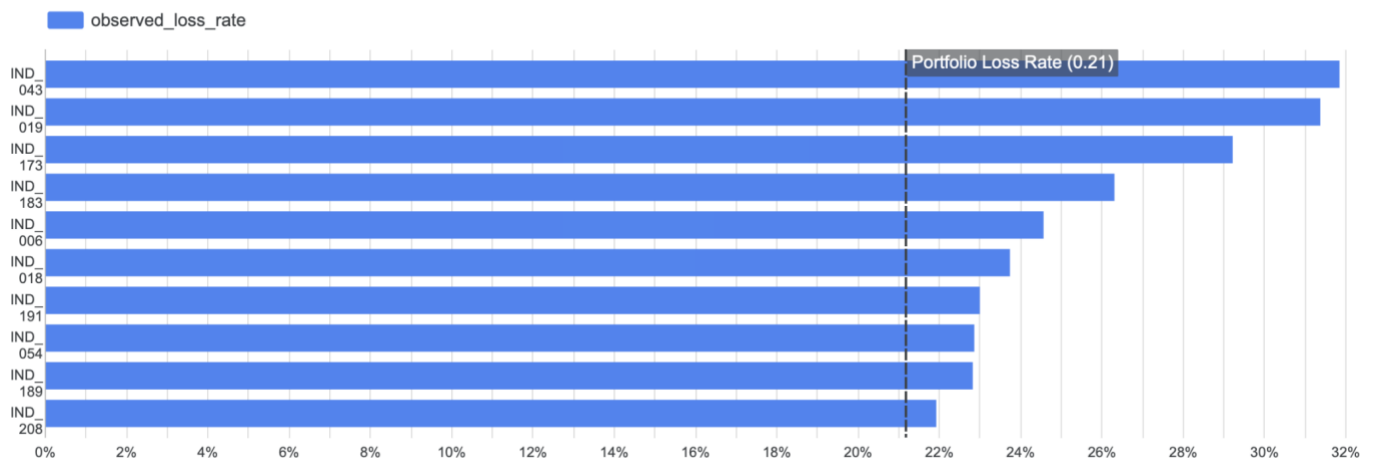
Finding 2 — Categorical segments separate risk, but frequency and severity differ

Feature screening

Funded Loan and Loss Loan Amount



Observed Loss Rate by Segment



Five categorical predictors were screened using the same loss metrics, with categories containing fewer than 100 loans excluded from rate comparisons to reduce instability from very small groups. feat_001 produced the widest loss-rate spread at 22.8 pp, but feat_003 produced a nearly identical 22.4-point spread with 55 categories rather than 232 and 26 categories meeting the 100-loan threshold versus 24 for feat_001. On this descriptive evidence, feat_003 offers a more compact candidate for modelling. feat_041 shows no monotonic pattern in either loss rate or severity and should be treated as nominal rather than ordinal.

Feature	Categor ies	Material ≥100	Loss rate range	Spread	Segm ent	Loan s	Loss rate	vs portfol io	Avg severity	Avg net loss / loan	Share of net loss
feat_001	232	24	9.1%–31.9%	22.8pp							
feat_003	55	26	12.6%–35.0%	22.4pp	IND_183	965	26.32%	+5.1pp	30,448	\$8,305	11.75%
feat_004	2	2	14.7%–24.9%	10.3pp	IND_019	411	31.39%	+10.2pp	52,155	\$16,722	9.99%
feat_015	2	2	18.2%–23.3%	5.2pp	IND_043	113	31.86%	+10.7pp	51,665	\$16,460	2.70%
feat_041	4	3	19.5%–22.6%	3.1pp	IND_173	195	29.23%	+8.0pp	25,343	\$7,663	2.17%

Figure 2. Loss frequency and severity by elevated feat_001 segment

Segment profile

Four feat_001 categories remain materially above the portfolio loss rate, but the economic pattern differs. IND_019 and IND_043 combine loss rates of 31.39% and 31.86% with average positive-loss severity of about \$52,000; their average net loss per funded loan is \$16,722 and \$16,460, around 2.4 times the portfolio baseline of \$6,853. IND_183 and IND_173 are primarily frequency-led: loss rates are 26.32% and 29.23%, while severity is at or below the portfolio positive-loss benchmark, producing average net loss per funded loan of \$ 8,305 and \$ 7,663. IND_183 accounts for 11.75% of net losses largely because it contains 965 loans, whereas IND_019 contributes 9.99% from 411 loans.

Vintage check and practical context

Vintage composition was examined because 2023 has the highest portfolio loss rate. IND_183, IND_019 and IND_173 have 2023 shares close to the portfolio's 24.5%, whereas IND_043 is more concentrated in 2023 at 66.4%. Within 2023, all four still exceed the portfolio's 26.71% loss rate: IND_019 42.42% (+15.7 pp), IND_043 40.00% (+13.3 pp), IND_183 33.57% (+6.9 pp) and IND_173 32.73% (+6.0 pp). The segment signal therefore persists within a common vintage, although IND_043's all-period gap is partly driven by its vintage mix.

These are descriptive, univariate associations rather than causal effects. feat_001 is also highly sparse: 208 of 232 categories contain fewer than 100 loans, and 115 contain fewer than 10, so extreme rates in tiny categories are statistically unstable; the sub-100-loan categories represent 34.8% of the portfolio and 36.0% of losses. Any modelling or policy use should therefore apply regularisation or category pooling and test whether the segment signal remains after controlling jointly for time and other predictors. Where the pattern is confirmed, severity-led segments would justify closer exposure and security review, while frequency-led segments would point more directly to underwriting or pricing.

Finding 3 — Positive losses are concentrated in a small tail

Statistical evidence

Positive realised losses are strongly right-skewed. Among 2,232 loans with loss_amount > 0, totalling \$68.91 million, the largest 224 cases account for 36.69% of positive losses; they are 10% of positive-loss cases but only 2.2% of all funded loans. The largest 20% of positive-loss cases account for 55.77%, and the Gini coefficient is 0.526. At the other end, 630 loans in the \$ 0–10k band represent 28.2% of positive-loss cases but only 4.6% of positive loss. The largest single recorded loss is \$ 440,332.

Top share of positive-loss loans	Loans	Share of positive losses
Top 1%	23	7.85%
Top 5%	112	23.45%
Top 10%	224	36.69%
Top 20%	447	55.77%

Severity band	Loans	% cases	% total loss	Mean	Median
0–10k	630	28.2%	4.6%	\$5,058	\$5,143
10k–25k	695	31.1%	16.7%	\$16,561	\$16,120
25k–50k	481	21.6%	24.4%	\$34,946	\$33,845
50k–100k	328	14.7%	32.8%	\$68,932	\$67,372
100k+	98	4.4%	21.5%	\$150,922	\$131,391

Figure 3. Positive-loss concentration by severity band or cumulative loss share

Limitations and practical context

This section deliberately uses loss_amount > 0 rather than is_loss = 1, so its population does not reconcile with the vintage and segment analyses. There are 2,232 positive-loss loans and 2,119 flagged loss loans, with 2,099 appearing in both; 153 loans disagree across the two outcome definitions. The tail is also recent: 14 of the 23 largest losses arise in 2024–2025, which account for 32% of funded loans. A portfolio average therefore understates concentration risk. Operational controls should focus on large individual loss exposures and early-warning review, while recognising that the current dataset does not contain facility size and therefore cannot distinguish large exposures from unusually poor recoveries.

Data Quality and Analytical Reliability

The following issues affect the reliability of descriptive estimates and should directly inform Part 2 validation.

- Potential duplicate structure. 5,427 rows (54.3%) have all 99 feature values identical to at least one other row, forming 2,435 groups; only 7,008 feature records are distinct. These groups also share received date, loss flag and loss amount, so portfolio rates are effectively weighted by repeated patterns and nominal loan counts overstate the independent information in the sample. A random train/test split could place identical patterns in both sets; at a 1 October 2024 time cut-off, no such groups straddle the split.
- Conflicting target definitions. 153 loans do not align across `is_loss` and `loss_amount`: 133 are flagged as no-loss but have a positive loss amount, up to \$ 101,341, while 20 are flagged as losses with zero or negative amounts. These conflicts should be resolved with the data owner or explicitly handled as separate target definitions; they do not support silently recoding one field from the other.
- Sentinel-coded missingness. Values such as 1e9, 99999 and 999 encode unobserved values in some `_min` and `_max` features. `feat_030_max` is 72.9% sentinel, and `feat_066_min` is 87.0%, so ordinary null checks materially understate missingness and affected summary statistics are not interpretable without recoding.
- Origination gap and non-stationarity. No funded loans appear in 2021 Q2–Q4, leaving only 152 loans in 2021, followed by a roughly sevenfold increase in volume in 2022. The cause is not available in the data. Combined with changing loss rates over time, this supports time-based rather than random validation.
- Other data issues. 197 loans have negative loss amounts, treated here as net recoveries; four loans have null loss amounts. Float32 storage is also present in the feature data, so derived continuous values should be compared using tolerances rather than exact equality.

Limitations and Next Steps

- Duplicate sensitivity. Recalculate core metrics on the 7,008 distinct feature records and compare them with the duplicate-weighted results. The size and direction of this sensitivity are currently unknown.
- Feature interpretability. Because predictors are anonymised, the business meaning of `feat_001`, `feat_003` and other drivers cannot be established from this dataset alone. `feat_003` is the more parsimonious categorical candidate for modelling, but production use would require a data dictionary and stability checks.
- Missingness. Convert sentinel values to nulls and retain explicit missingness indicators before modelling; otherwise summary statistics and model splits can treat missing codes as real extremes.
- Target governance. Resolve the 153 outcome conflicts before final model sign-off and define clearly whether Part 2 is predicting the supplied `is_loss` flag, positive realised loss, or a harmonised credit-loss definition.
- Exposure and recovery gap. The data do not contain original balance, exposure at default, collateral value or recovery timing, so the analysis cannot determine whether the rise in severity reflects larger facilities or worse recovery performance.
- Validation and modelling. Use time-based validation because the portfolio mix and loss rate change over time and 2021 contains an origination gap. For Part 2, modelling frequency and severity separately, or modelling net loss directly, is more aligned with the observed divergence between default incidence and loss size than a loss-flag-only model.